University 5.0: Powerful Agile Learner

Pratikno

10 Februari 2018

Industrial Rev 4.0 & University 5.0

(Pratikno, 2018)

INDUSTRIAL REVOLUTION 4.0
(Digitization, Computing Power, Data Analytics)

Technology:
- Cyber-physical
- Internet of Things
- Biology

Application:
- Agricultural
- Logistics
- Manufacturing
- Etc

Content

DISRUPTION:
(Uncertain, Unpredictable Future; Constant Change)

The Deep Shift:
- Physical → Virtual
- Human → Automation & Robotics
- Intermediary → Disintermediary

The Paradox:
- Voice vs Surveillance
- Empowerment vs Agglomeration
- Jobs Lost vs Jobs Gained
- Abundance Scarcity

Content

BE THE CHAMPION:
(Disruptor instead of disrupted, taking benefit from unpredictable future; Drive the future)

Organiz Ecosystem:
- Flexibility & Agile
- Talent Pooling
- Goals oriented
- Collaboration & Sharing

Powerful Agile Learner:
- Empathy
- Curiosity & Creativity
- Complex Problem Solving
- Determination

Character

Multi→Inter→Trans-Disciplinary

… INDUSTRIAL REVOLUTION 4.0 …

Optimizing the existing knowledges & technologies through Digitization, Computing Power & Data Analytics

A

***Cyber-Physical*:**

- Autonomous Vehicle
- 3D Printing
- Advanced Robotics
- New Materials

B

Internet of Things:

- Wearables
- Big Data, AI
- Block chain & Cryptocurrency
- Augmented/Virtual Reality

C

Bio-Technology:

- Gene Sequencing/ Edit DNA
- Synthetic Biology
- Personalized medical treatments
- “New” Neuroscience

Cyber Physical System

A. Cyber Physical





Maturation of new cyber physical technologies (artificial intelligence, 3-D printing, robotics)

Data analytics driving efficacy and effectiveness and new business models

Pervasive sensing and actuation

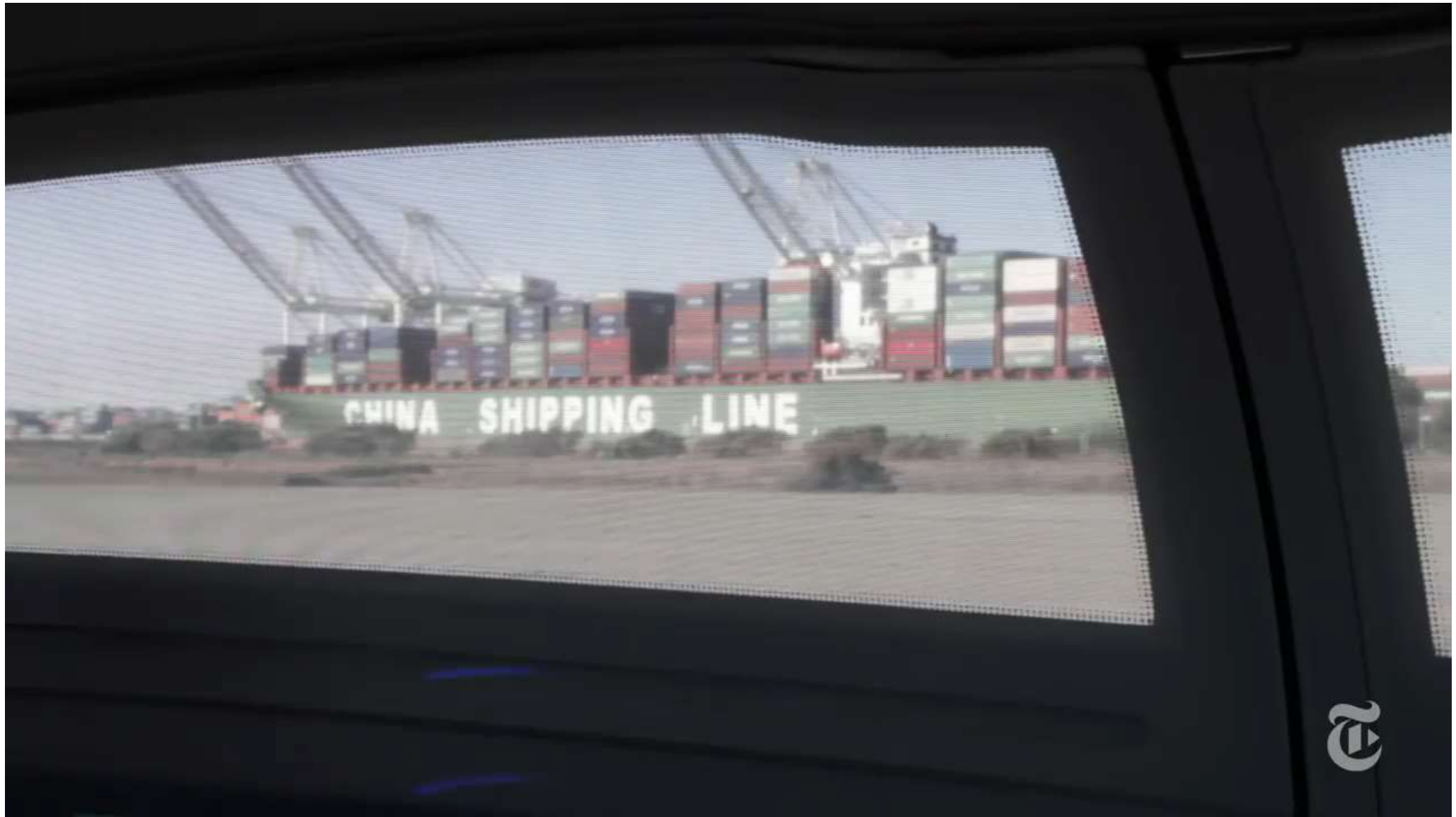
Ubiquitous connectivity throughout the supply chain

Unprecedented levels of data and increased computing powers

McKinsey&Company | Source: *Forbes*; World Economic Forum

A. Cyber Physical

Autonomous Vehicle



A. Cyber Physical

3D Printing



It only takes 24hrs to build

A. Cyber Physical

Advanced Robotics



Robots do 70% of work

A. Cyber Physical

High Tech Sustainable Farming

The *Tiny* Netherland: 2nd largest agricultural products exporter

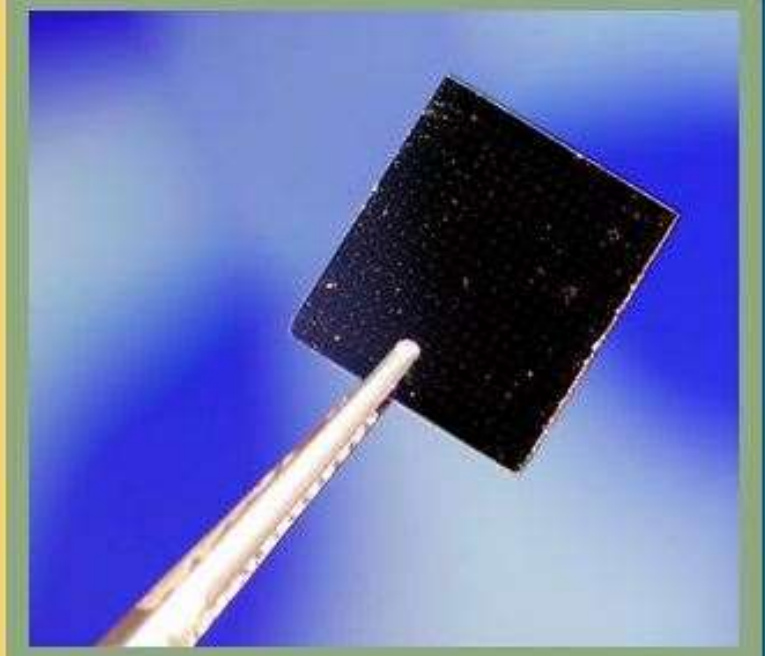
Country	Exports, Billion $US				% of world total			
	1995	2005	2010	2016	1995	2005	2010	2016
World	361.2	538.6	870.3	1,054.7	100.0	100.0	100.0	100.0
United States	42.2	48.3	82.6	99.9	11.7	9.0	9.5	9.5
Netherlands	34.1	42.7	64.5	71.5	9.4	7.9	7.4	6.8
Germany	21.2	35.2	53.0	59.8	5.9	6.5	6.1	5.7
China	10.0	22.5	41.1	56.8	2.8	4.2	4.7	5.4
Brazil	10.0	21.8	45.2	45.0	2.8	4.0	5.2	4.3
France	30.3	32.9	43.8	41.8	8.4	6.1	5.0	4.0
Spain	11.5	21.8	29.9	38.4	3.2	4.1	3.4	3.6
Canada	11.7	20.7	28.5	35.6	3.2	3.8	3.3	3.4
Belgium	11.5	23.6	31.3	34.0	3.2	4.4	3.6	3.2
Italy	11.2	17.0	25.6	29.2	3.1	3.2	2.9	2.8
Thailand	10.7	12.4	23.9	26.8	3.0	2.3	2.8	2.5



A. Cyber Physical

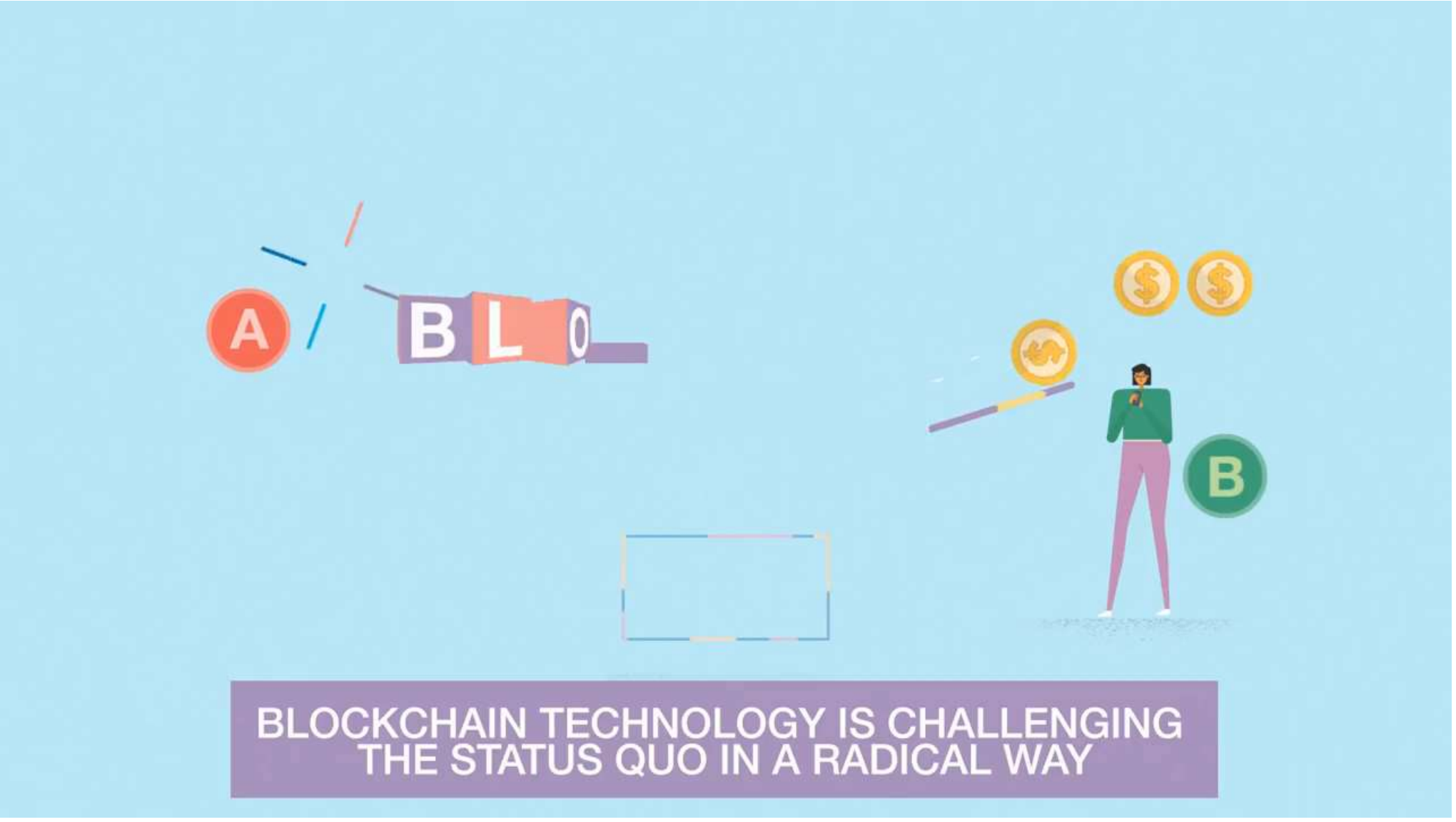
New Materials: Graphene

GRAPHENE, THE WORLD'S STRONGEST MATERIAL, IS A MILLION TIMES THINNER THAN PAPER BUT 200 TIMES STRONGER THAN STEEL.



B. Internet of Things

Blockchain & Cryptocurrency

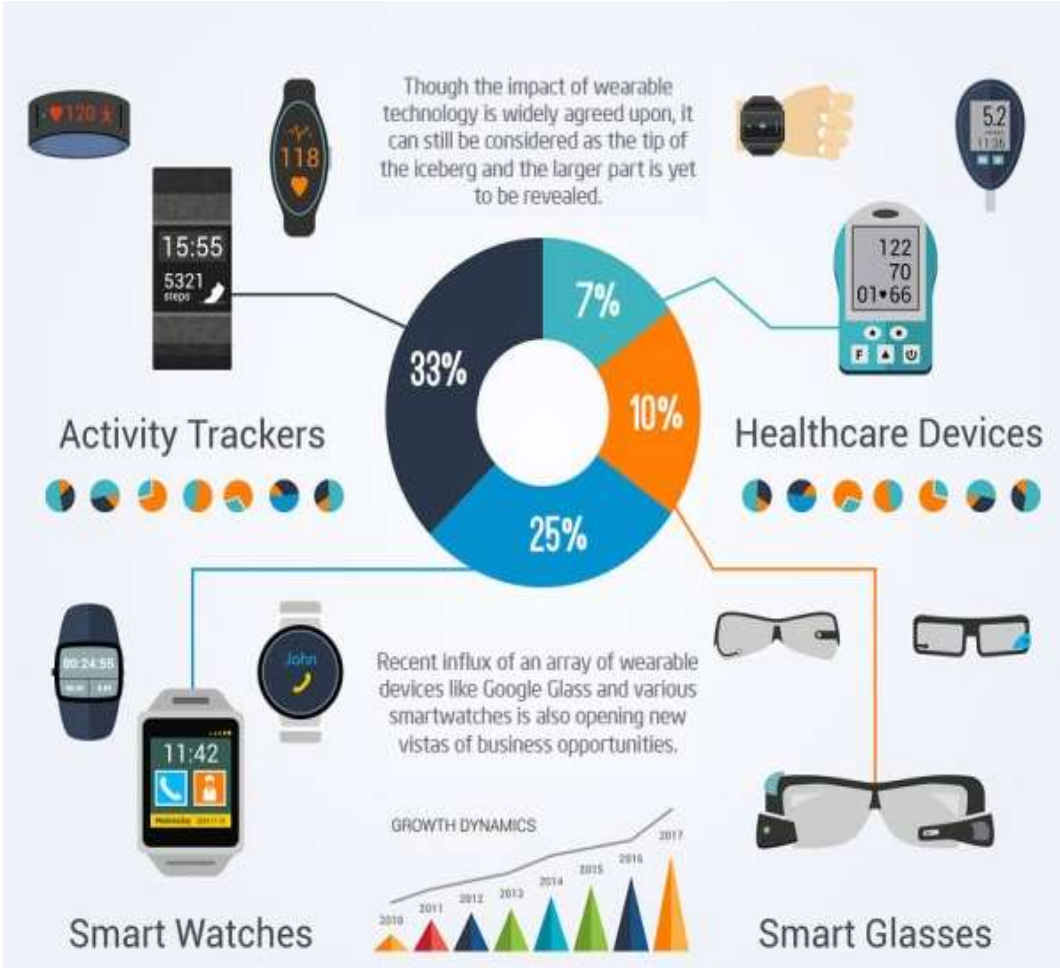


Disintermediacy of disintermediation

B. Internet of Things

Wearables





B. Internet of Things

"Who owns the information, owns the world", Rothschilds



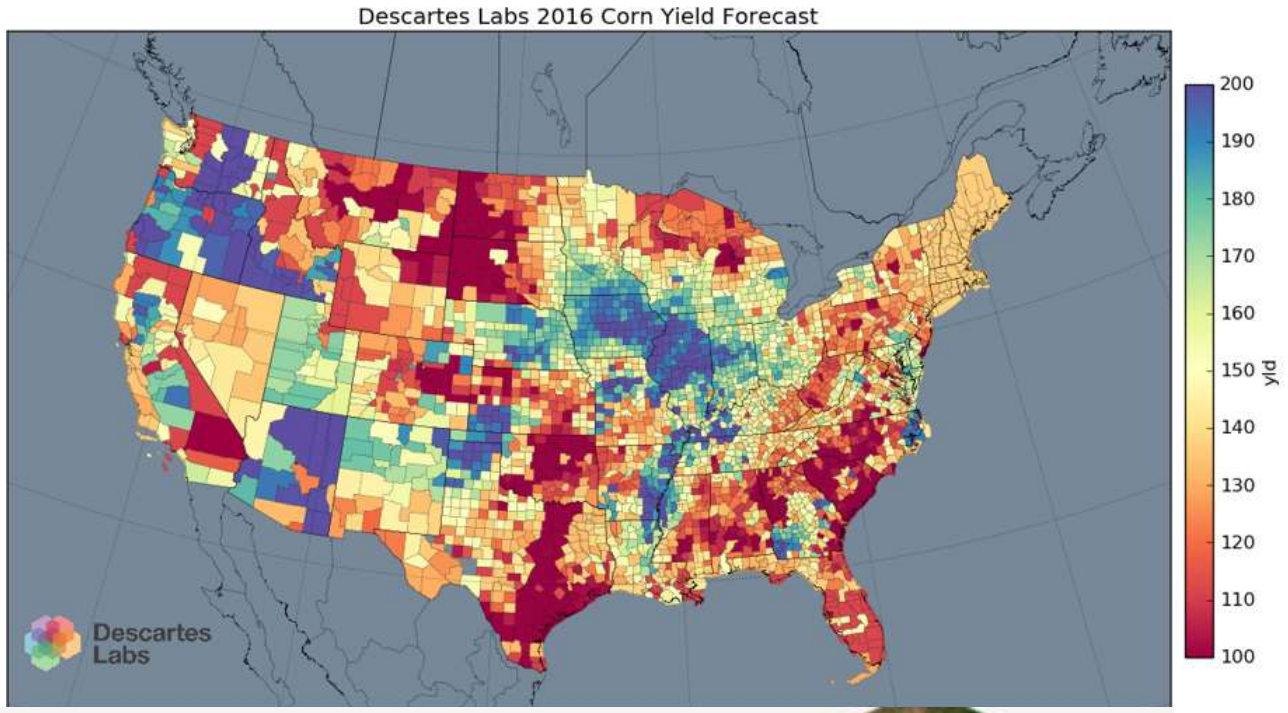
"The world most valuable resource is no longer oil, but DATA"

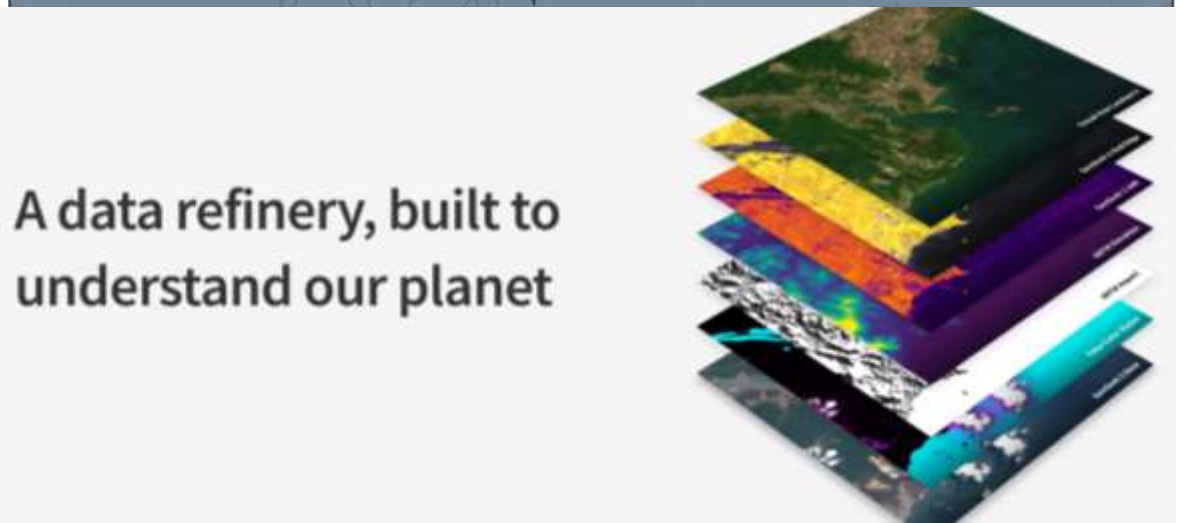
Data Analytics



B. Internet of Things

Big Data & Machine Learning





Descartes Labs

Startup Descartes Labs:

- **prediksi hasil panen jagung per hari, tidak lagi per bulan**
- **lebih akurat + efisien, bahkan dibandingkan US Department of Agriculture**
- **Manfaatkan big data citra satelit, algoritma,AI**

B. Internet of Things

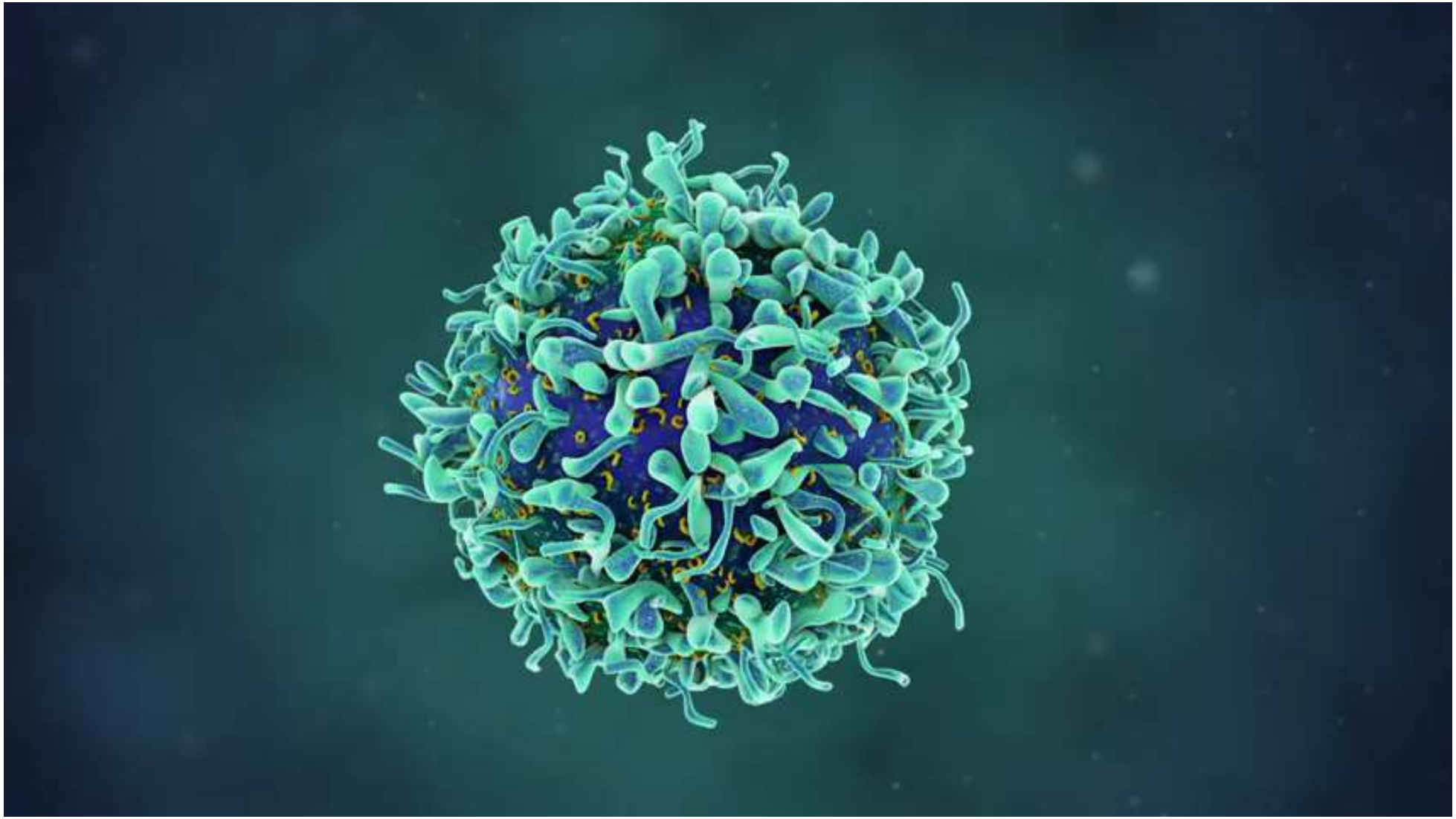
Artificial Intelligence



Sophia,

C. Biotechnology

Personalized Medical Treatments



More effective, precise medicine for individual

C. Biotechnology

Edit DNA & Synthetic Life

LIFE CODE:

C. Biotechnology

Edit DNA & Cyber Physical



1 sapi
1100 kg
susu bubuk
per tahun

Industrial Revolution 4.0:

Disrupt the Existing Structure

Huge benefit !!



Huge problems !!

Expand the creative disruption;
disrupt the destructive one!!

… DISRUPTION …

Constant Change, Uncertain & Unpredictable Future

The Deep Shift:

- Physical → Virtual
- Human → Automation and Robotic
- Intermediacy → Disintermediacy

The Paradox:

- Voice vs Surveillance
- Empowerment vs Conglomeration
- Jobs Lost vs Jobs Gained
- Scarcity, but Abundance

Disrupt the Existing Businesses



BukaLapak.com
Belanja dimana, aman dan terpercaya!

blibli.com
BIG CHOICES BIG DEALS

tokopedia

VS



NETFLIX

Disrupt the Existing Businesses

facebook

The biggest social media, does not produce content

UBER

The biggest taxi company, does not have cars

Alibaba.com
Global trade starts here.™

The biggest retailer, does not have its own stocks

airbnb

The biggest accommodation service, does not have properties







Even Banking too !!

Disrupt the Existing Jobs

















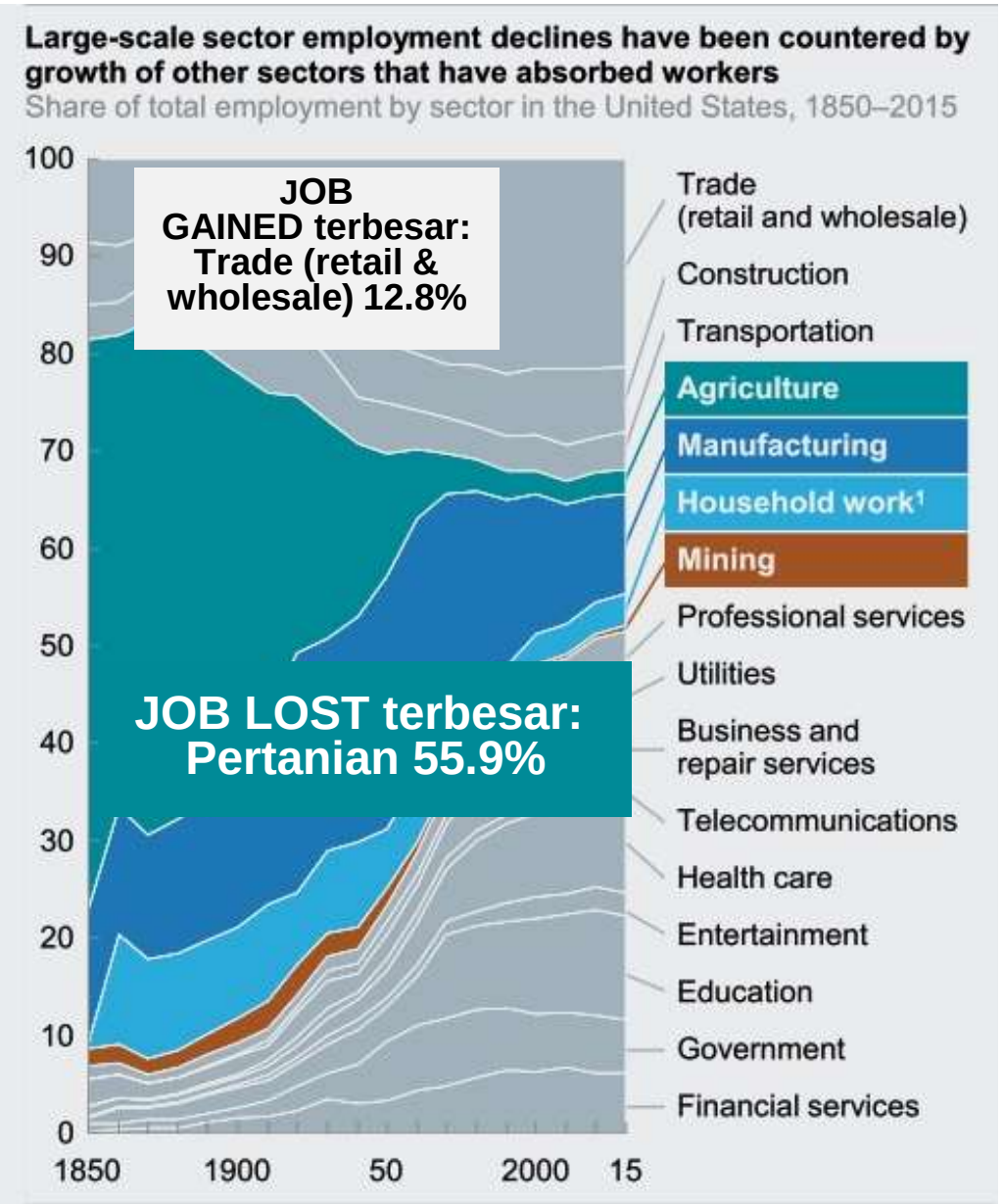


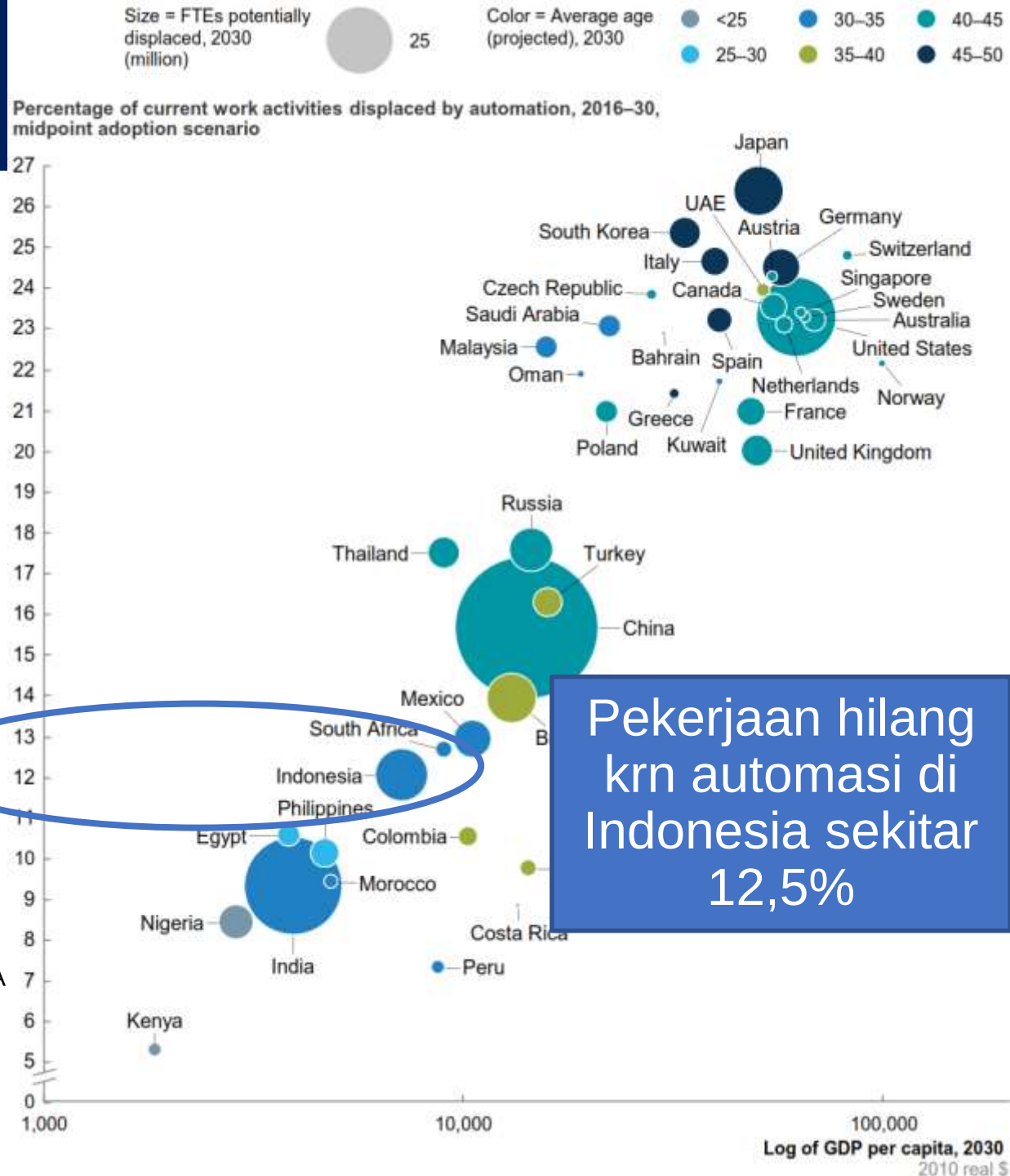



1. Farmers and ranchers
2. Postal mail staff
3. Sewing machine operators
4. Switchboard operators
5. Fast food cooks
6. Agricultural workers
7. Data entry keyers
8. Typists
9. News street vendors
10. Food service managers
11. Electrical assemblers
12. File clerks
13. Pre press technician
14. Computer operators
15. Post masters
16. Office machine operators
17. Textile and garment
18. florists
19. Pump and refinery operato
20. Loan interviewer/clerks

Source: https://www.forbes.com/pictures/lmj45ighg/top-20-disappearing-jobs/#49c61e624bc5

Disrupt the Existing Jobs

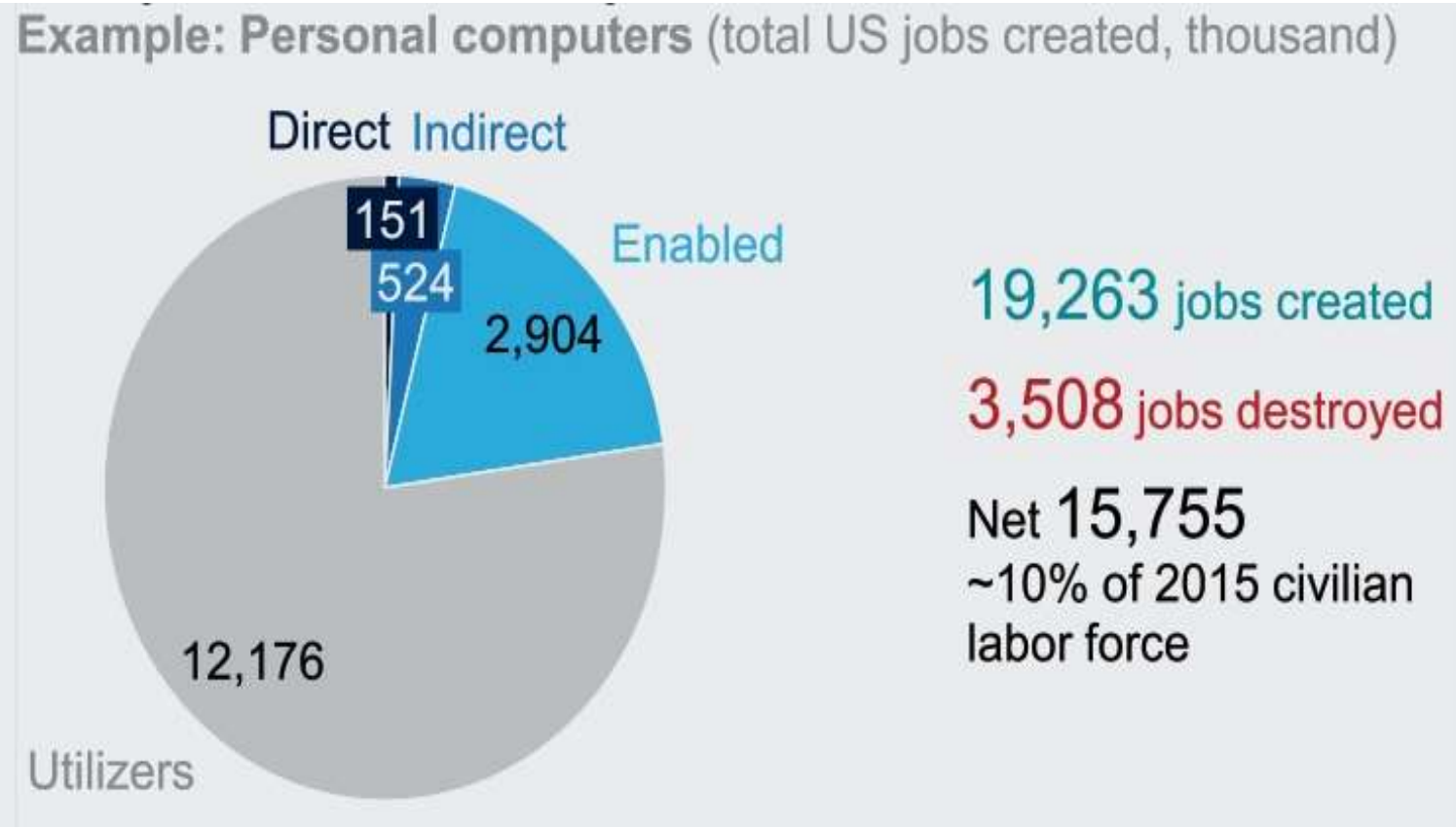




Pekerjaan hilang krn automasi di Indonesia sekitar 12,5%

Source: IPUMS USA 2017; US Bureau of Labor Statistics;; McKinsey Global Institute analysis , McKinsey 2017

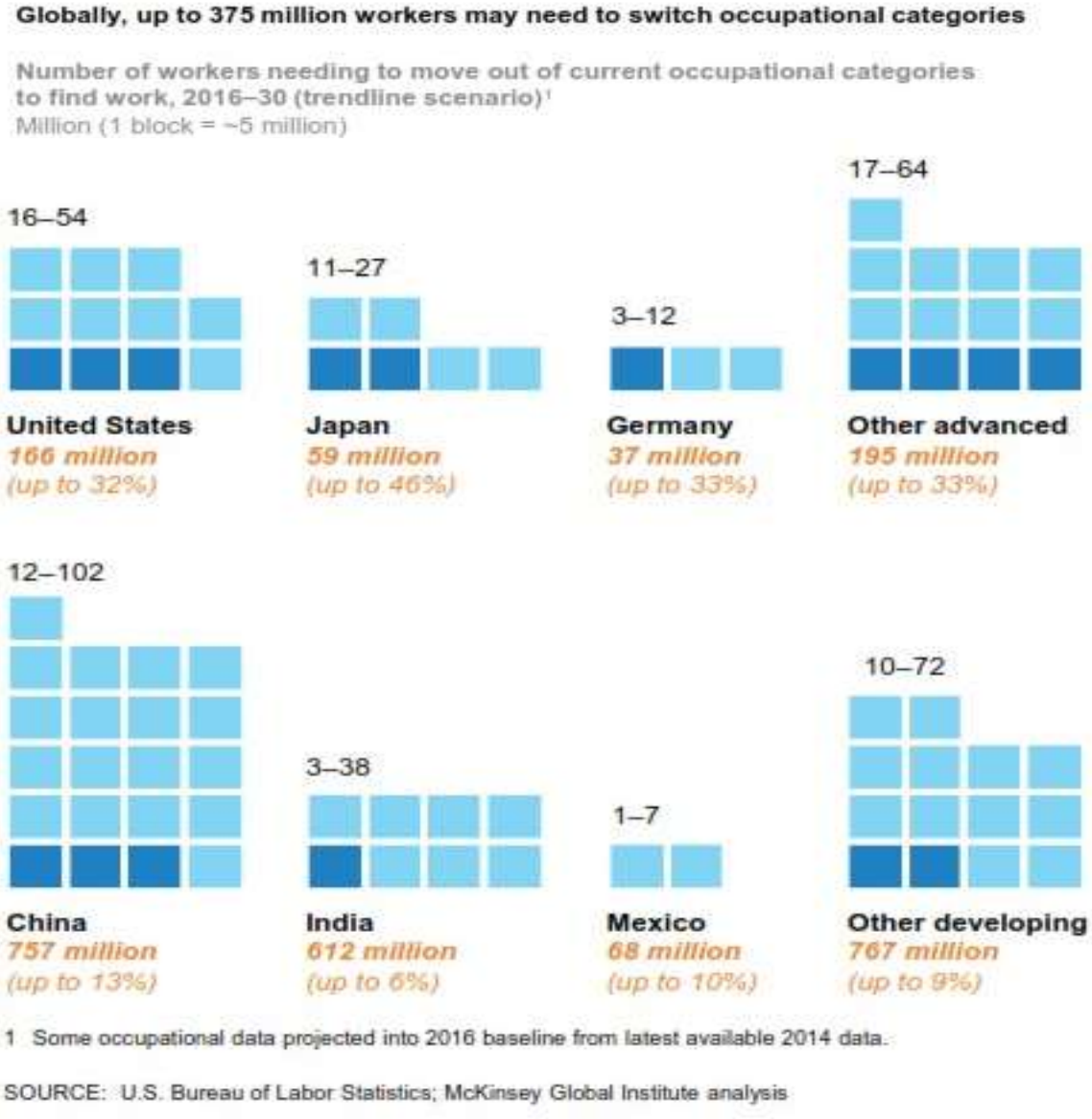
Disrupt the Existing Jobs

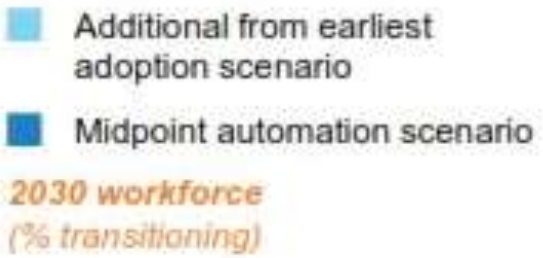


Technology... creates MORE JOBS, than it destroys.. mainly OUTSIDE the industry itself

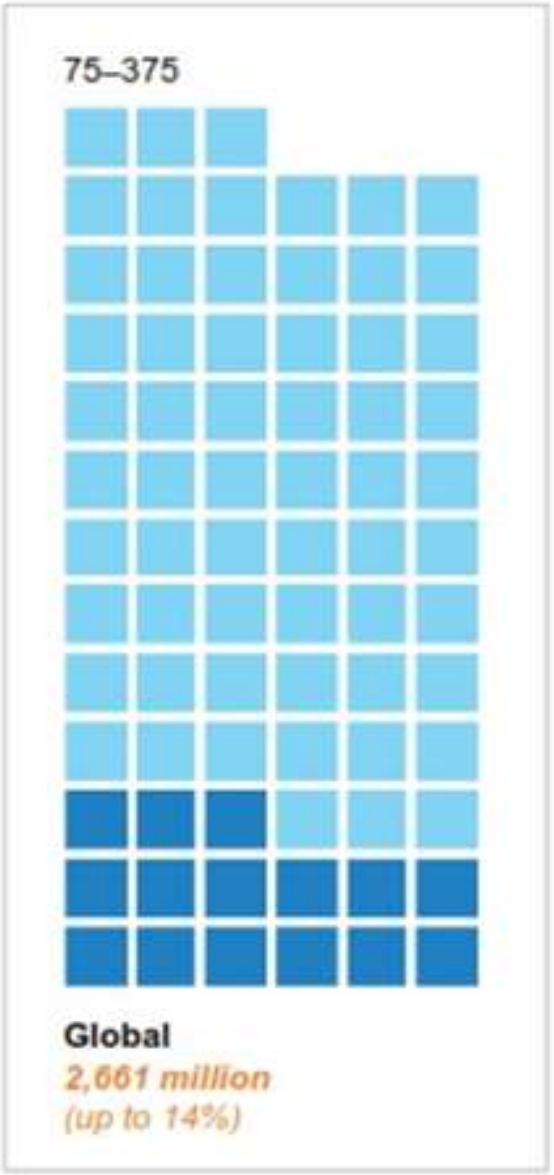
Source: IPUMS USA 2017; US Bureau of Labor Statistics; Groningen Growth and Development Centre 10-Sector Database; Moody's; IMPLAN; US Bureau of Labor Statistics; FRED; McKinsey Global Institute analysis

Disrupt the Existing Jobs

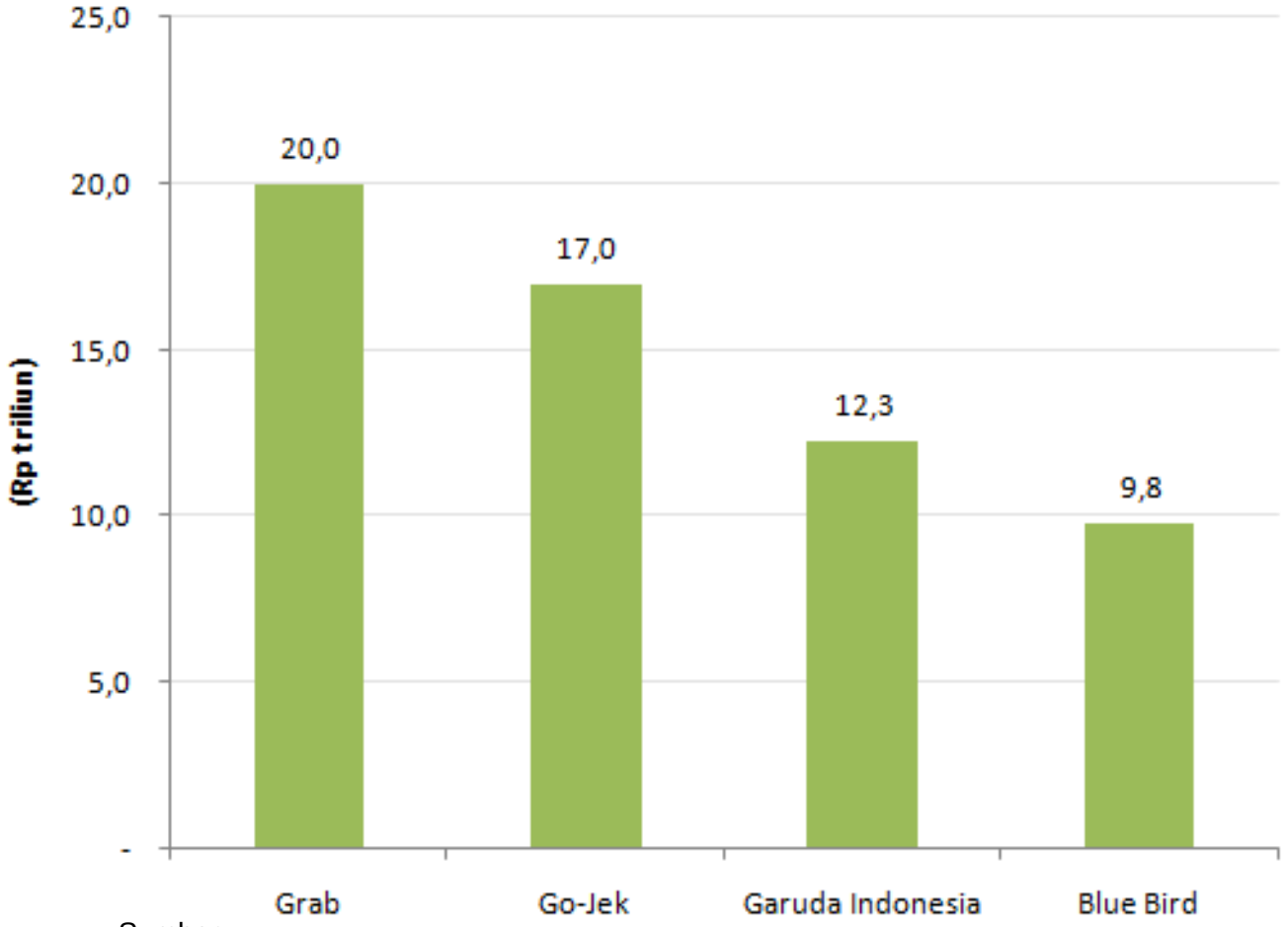




75 to 375 million workers need to SWITCH occupational categories and LEARN new skills



The Deep Shift: Heavy Asset to Light Asset



Sumber:
http://media.bareksa.com/resources/image/2016/08/13779/id_14709046151.png

Valuasi Gojek tahun 2016 Rp17 triliun
LEBIH BESAR
dari Garuda Indonesia dan Blue Bird.

Kini, di tahun 2018 valuasi Gojek sudah mencapai sekitar Rp40 triliun



NOW AVAILABLE ON GO FOOD POWERED BY GO-JEK
GO MASSAGE POWERED BY GO-JEK
GO GLAM POWERED BY GO-JEK
GO TIX POWERED BY GO-JEK
GO PULSA
GO CLEAN POWERED BY GO-JEK
GO-SEND
GO MART POWERED BY GO-JEK

THE LARGEST COMPANIES BY MARKET CAP

The oil barons have been replaced by the whiz kids of Silicon Valley

The Deep Shift



visualcapitalist.com

The Deep Shift:
Big Data Analytics, Machine Learning, AI...



Gojek: Big Data, Machine Learning, AI, Alghorithm

- **Menganalisis dampak larangan motor di Sudirman: cepat, efisien → normalnya perlu survey beberapa bulan**
- **Membuat proyeksi**
- **Bisa untuk laporan cuaca, real time**
- **Bahkan risk profiling**
- **dll**

The Deep Shift: Physical to Virtual











The Deep Shift: Physical to Virtual





The Deep Shift:
Intermediacy to Disintermediacy

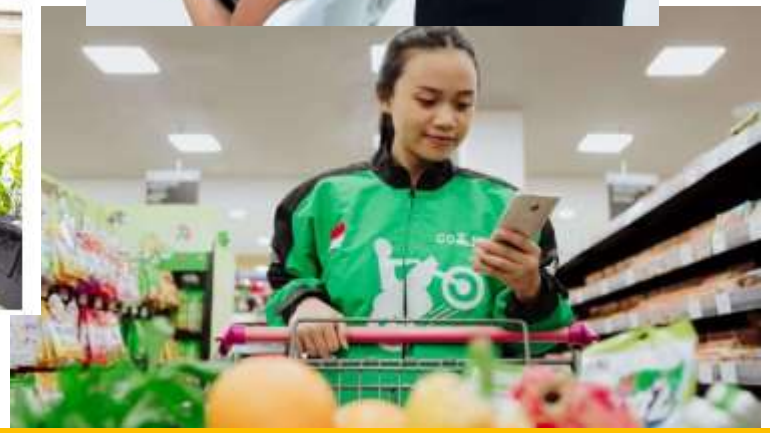














Disintermediasi, on-demand services, immediacy, 24/7

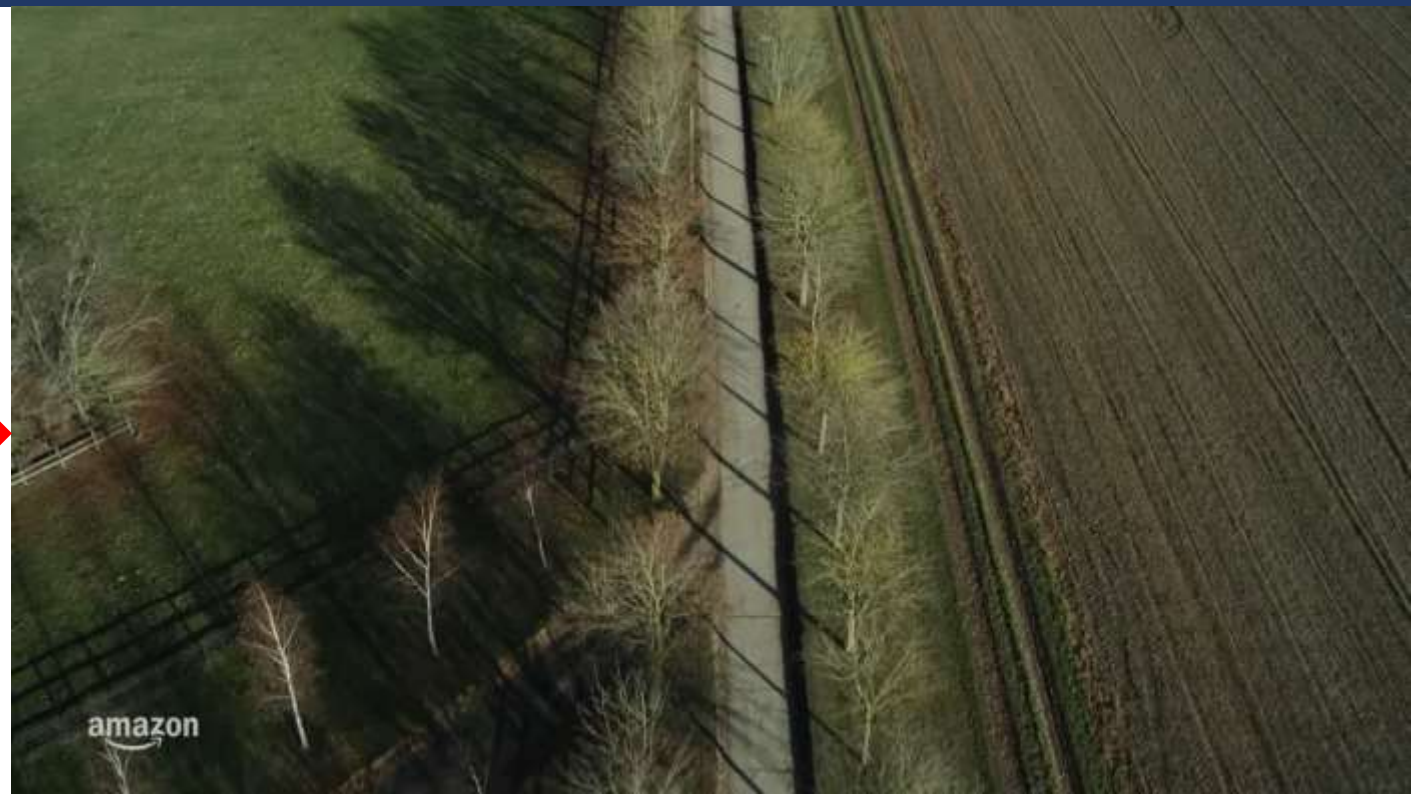
The Deep Shift: Virtual Office



The Deep Shift:
Human to Automation & Robotics



Kurir Logistik



Amazon Prime Air:
Actual Try Out of Drone Delivery

The Deep Shift:
Human to Automation & Robotics



Tukang Bangunan



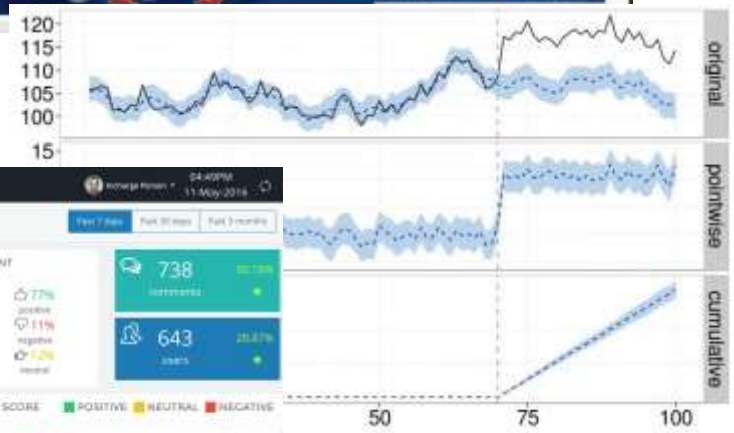
3D printing devices
misal, membangun jembatan

The Deep Shift:
Big Data Analytics, Machine Learning, AI...

For Government:

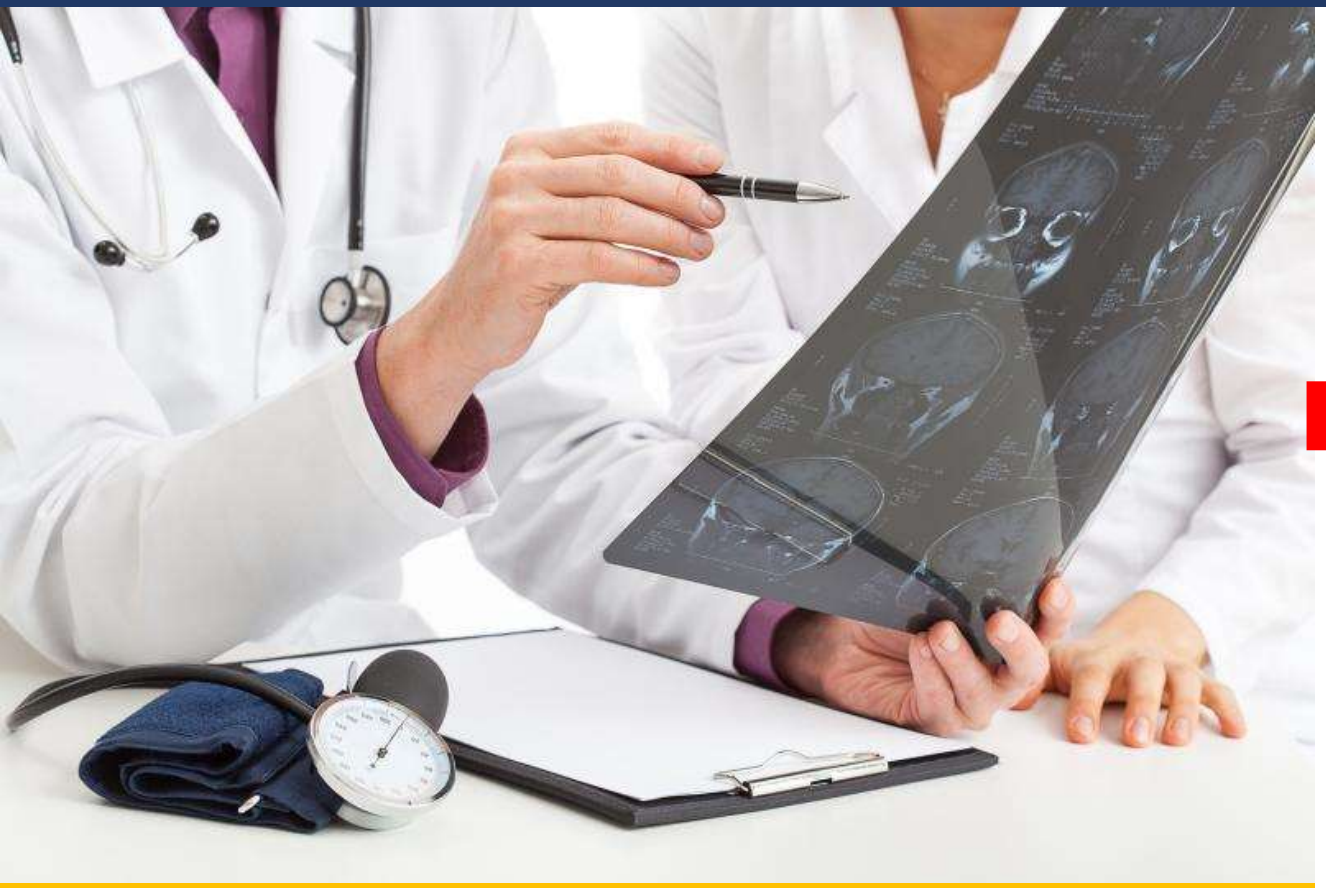
1. **Sentiment Analysis**
   - Citizen aspirations
   - Public policy
   - Accurate infrastructure development
2. **Causal – Impact Analysis**
   - Campaign vs effect (real time)
   - National-events impact analysis
3. **Dashboard & Monitoring**
   - Manajemen asset dengan IoT
   - Laju ekonomi antar daerah



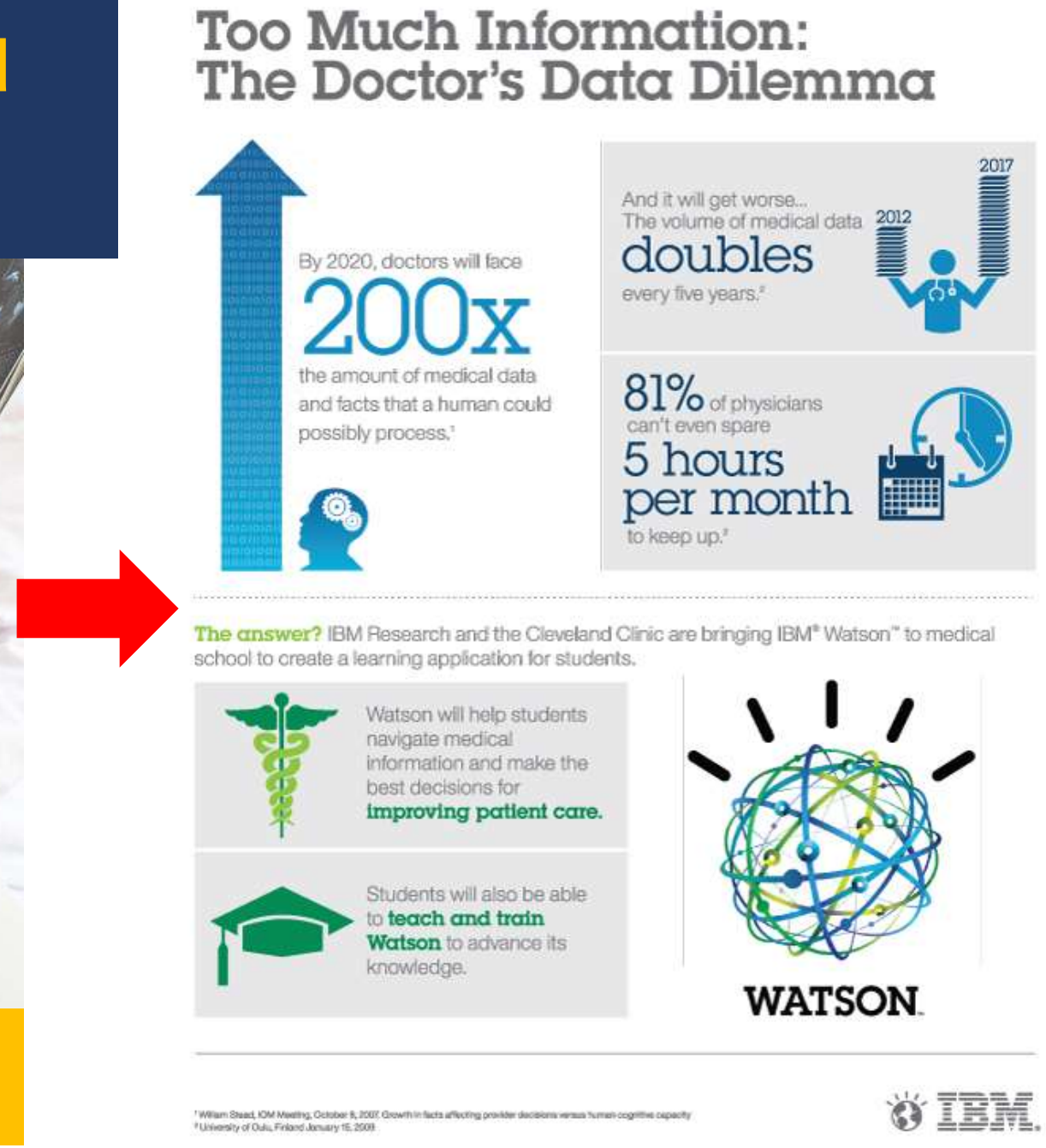




The Deep Shift: Personalized Medical Treatments

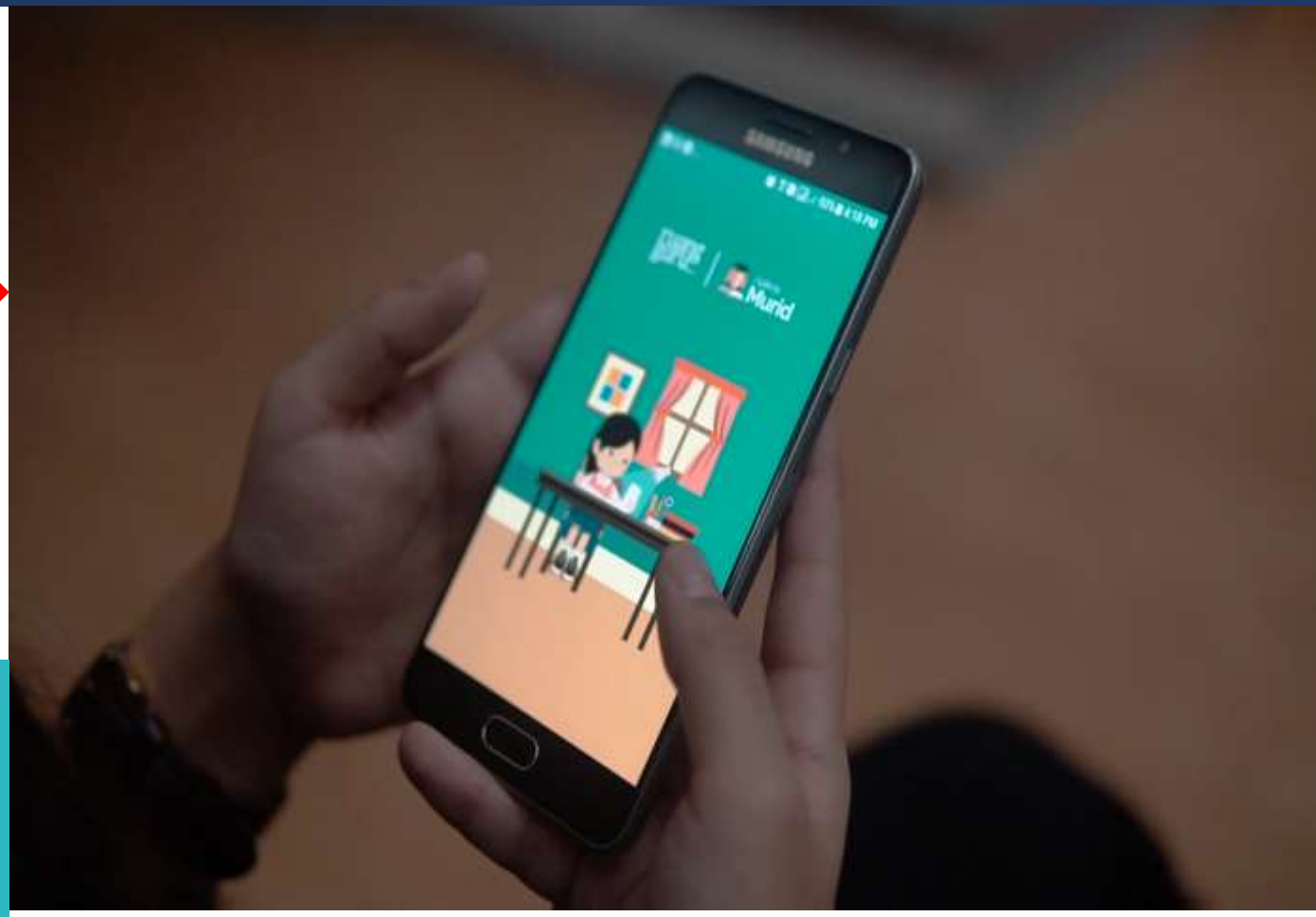


Diagnosis Dokter



The Deep Shift:
Learning Environment, Methods, & Delivery





ruang guru

Biggest learning apps in Indonesia with 6 million users

Tech based education services provider:
Virtual class
Online test, learning, & tutoring
Monitoring Dashboard

The Deep Shift:
Learning Environment, Methods, & Delivery

coursera

UDACITY

edX
Take great courses from the world's best universities

IndonesiaX
Enriching Lives Through Education

Massive Open Online Courses/MOOC:

1. kursus dan kuliah *online* gratis (sebagian bayar murah)
2. terbitkan sertifikat yang mulai diakui industri
3. akses kapan saja, tanpa harus hadir

Makin diminati:

1. Coursera → 2,7 juta siswa terdaftar, 50-100ribu siswa mengakses materi dari professor terbaik dunia;
2. Udacity → salah satu kursus 'Bahasa pemrograman' diikuti 200ribu siswa dari berbagai dunia

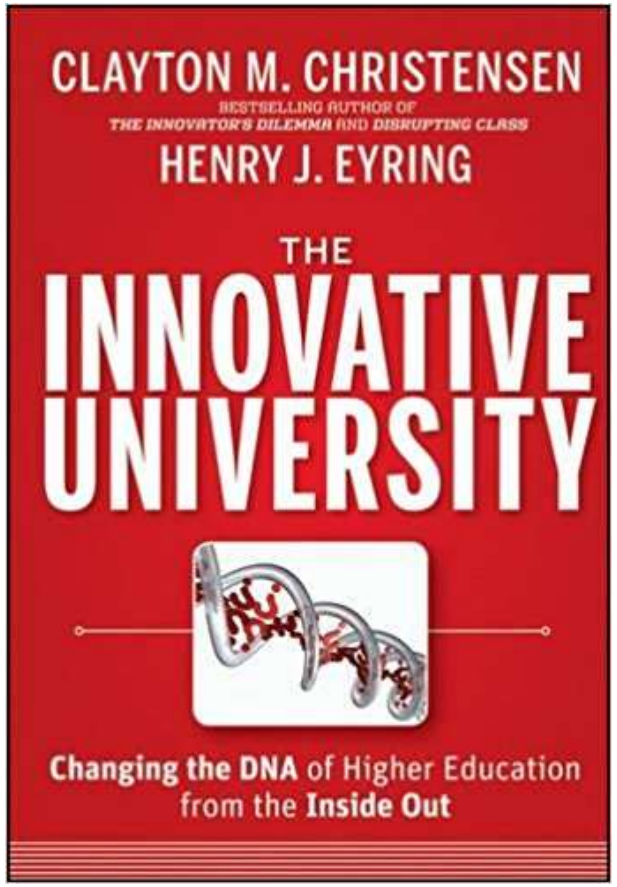
Perush besar mulai tidak bertanya ijazah, melainkan: *portfolio, competency based*

Google

ERNST & YOUNG

Perushaan ternama dunia semakin mengakui lulusan MOOC
Bahkan Harvard, MIT, Stanford, Princeton, dll telah bergabung…

The Deep Shift:
Learning Environment, Methods, & Delivery



Christensen & Eyring, 2011

Univ kecil kalah nama, tapi:

- Inovatif, *online learning*
- Lebih efisien, hanya US$10,000 sampai lulus
- Ceruk pasar: lintas kelas ekonomi, tawarkan fleksibilitas (waktu, desain kurikulum, dll)
- Beroperasi sepanjang tahun & waktu (24/7)

THE PARADOX

The Paradox: Empowerment ‘vs’ Agglomeration



Indonesia

East Asia & Pacific		Lower middle income	
Population, age 15+ (millions)	177.7	GNI per capita ($)	3,580

	Country data	East Asia & Pacific	Lower middle income
Account (% age 15+)			
All adults	36.1	69.0	42.7
		67.0	36.3
Adults belonging to the poorest 40%	22.2	60.9	33.2
Young adults (% ages 15–24)	35.2	60.7	34.7
Adults living in rural areas	28.7	64.5	40.0
Financial Institution Account (% age 15+)			
All adults	35.9	68.8	41.8
All adults, 2011	19.8	55.1	28.7
Mobile Account (% age 15+)			
All adults	0.4	0.4	2.5

- Industri transportasi online: **sumbang 500ribu Naker** (BPS, 2016)
- Banking the unbanked through digital financial inclusion – **around 64%** Indonesia population is **unbanked** (World Bank, 2014)

Google

VS

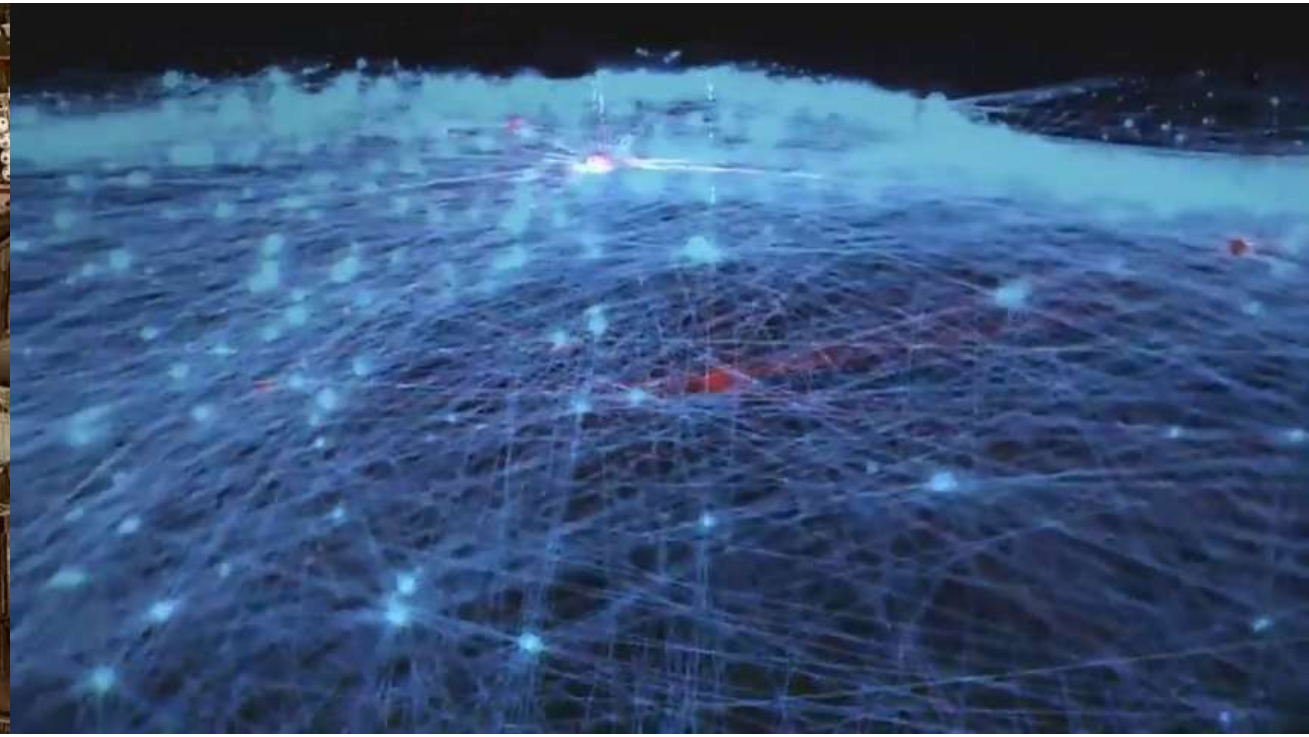
facebook

amazon

Alibaba.com
Global trade starts here.™

The Paradox:
Voice vs Massive Surveillance & Control





China: All Seeing State…

Middle East: Weapon of Mass Surveillance…

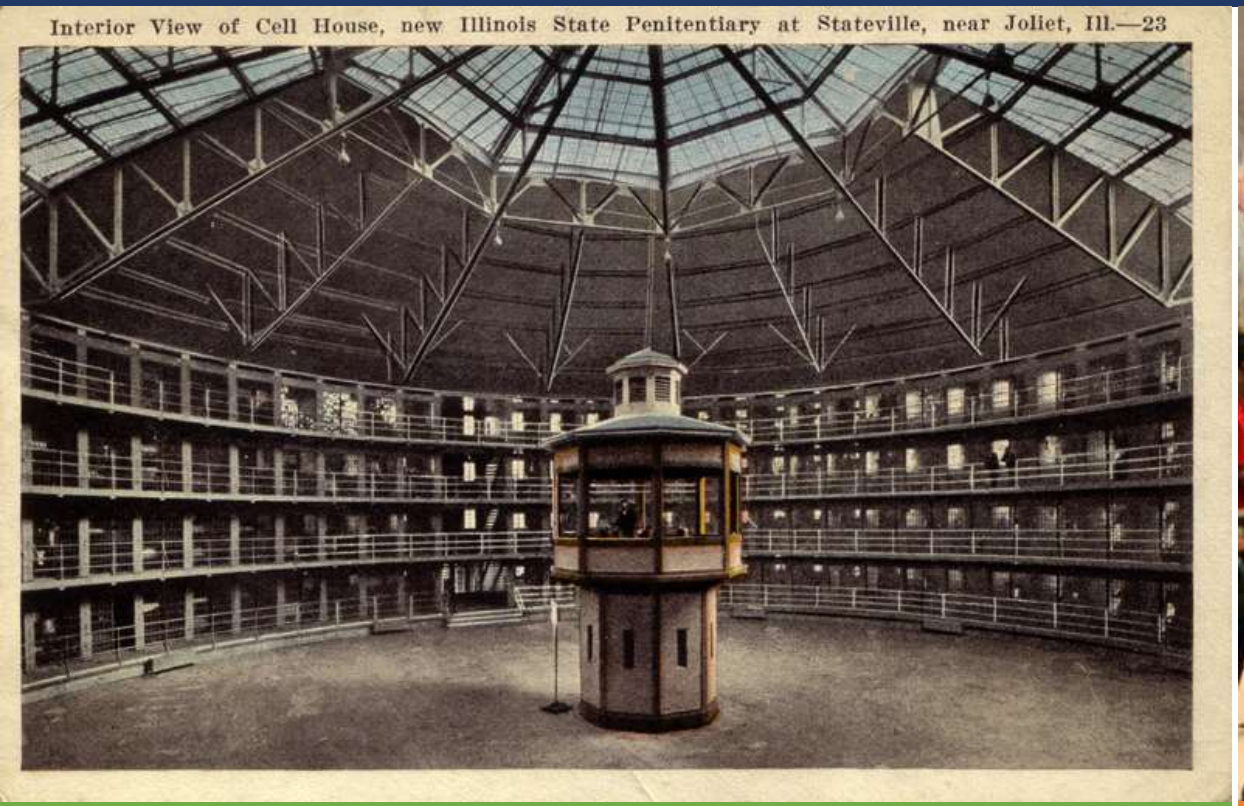
The Paradox:
Voice vs Massive Surveillance & Control





US NSA' PRISM & UK GCHQ' TEMPORA:
Sees everything, hears everything…

The Paradox:
Panopticon Society



‘Old’ Panopticon:
Penjara di Illinois, Stateville



‘Hyper-modern’ Panopticon Office

Foucault dalam Discipline & Punish (1985): Panopticon hadir untuk mendisiplinkan masyarakat, konteks dahulu untuk penjara, RSJ, sekolah, dll. Konteks masa kini: *mass surveillance…*

The Paradox:
Future Concern on Swarm/
Battle Drones

Technology facilitates !!

However, the recent implications: it just HAPPENED, it has HAPPENED…

SEARCH

The New York Times



Pence and Jordan's King 'Agree to Disagree' on Jerusalem



THE INTERPRETER
Trump's Hard-Line Israel Position Exports U.S. Culture War Abroad

Deadly Siege at Kabul's Largest Hotel



MIDDLE EAST

Russia Says Its Syria Bases Beat Back an Attack by 13 Drones

By NEIL MacFARQUHAR JAN. 8, 2018

MOSCOW — The main Russian air and naval
attack by 13 armed drones over the weekend,
Defense said on Monday.

The small drones were shot down before they
were no casualties, according to a statement
page.

But the attack could indicate that militants o
President Bashar al-Assad were stepping up
the Russian military, which intervened in Sep
beleaguered Syrian government, then its mai

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#INDUSTRIALS JANUARY 11, 2018 / 8:16 PM / 11 DAYS AGO

Russia investigating drone attacks on yria bases - RIA

ers Staff

1 MIN READ

SCOW, Jan 11 (Reuters) - Russia's defence ministry said on Thursday it was investigati
ch country produced explosives used in recent drone attacks on Russian bases in Syria,
news agency cited the ministry as saying.

e explosive material in question is produced in a variety of countries, including in
aine," RIA quoted major general Alexander Novikov as saying. (Writing by Polina Ivano
ing by Vladimir Soldatkin)

Standards: The Thomson Reuters Trust Principles.

most popular



Government shutdo
Trump attacks Dems
and calls for 'nuclear



Live Minnesota Vikin
Philadelphia Eagles:
championship game



Gareth Bale given st
ovation as Real Madr
Deportivo



Milford Track: 'Work
walk' in danger of be
just another bucket l



Sarah Ferguson clain
in damages from Mu
News Group

a job Sign in Search ~ International ed

The Guardian

on Sport Culture Lifestyle More ~

icas Asia Australia Middle East Africa Inequality Cities Global development

Russia implies US involvement in drone strikes on Syria military bases

Attacks on its bases in Syria would have required assistance from a country with satellite navigation, says Russian defense ministry, a charge the Pentagon denies



Russia has said that a recent series of drone attacks on its military bases in Syria would have required assistance from a country possessing satellite navigation

The Paradox: Concern of Defense & Security



https://www.youtube.com/watch?v=SGHOFxeaDr4

3D Printing & the Proliferation of Guns

The Paradox: Disruptive Social Behavior



Non-affiliations
No Command Structure
No Support from any group
Could happens anytime, anywhere
Motivated by ideology or belief through multi channels: self-exposed

What's next?



Let's Redefine:

GOALS



STRATEGY



PEOPLE



PROCESS

STRUCTURE

TECHNOLOGY







Flexibility & Agility of Organization

Understanding the Main Users (Students & Lecturers):

- Identify users needs,
- Define experience principles
- Bringing empathy → solving the problems
- **Streamlining process & Debureaucratization**

Incorporating Technological Advancement:

- Assess current technology environment
- Assess tech, process, organizational needs for realization
- Create tech-dev (agile) plan
- Bringing empathy
- **Digitization**

Designing in Real Time:

- Setting up war room
- Deploy small groups from multi-backgrounds (engineering, research, user experience, industrial design, etc)

Acting Quickly:

- Rapid prototyping
- Better learn based on real users feedback
- Frequent adjustment

Transformation at all Level

Level	Transformational Modes
Students	• Personalized learning experiences • Support & feedback • Recognition of achievement (co & extra-curricular)
Lecturers	• Targeted instruction: data-driven • Recognition of achievement (publication, hilirisasi riset)
Schools & Faculties	• Transform curriculum (student-centered, problem based) & delivery method
University	• Transform physical-virtual teaching & learning spaces, • connecting the dots (Innovation ecosystem)

EDUCATION

Orchestration

Learning Activities

Assessment Activities

On-Campus

Flipped classroom, Creative hub/ collaboration rooms, self-study spaces, library, labs

Off-Campus

Profesional placements (internship), co & extra curricula (student activism, etc), informal setting (community)

Virtual & Online

3D virtual worlds, digital content, simulations, etc → Platform seperti RuangGuru, Indonesia X/ MOOC, UGM eLOK/eLISA, SPADA Kemristekdikti

Tools, resources, people, environments

PERSONALIZED EDUCATION



Personalized & different paths for learner: Same Goals (Christensen, 2014)

Integrated Digital Content

Digital content allows for a differentiated path and pace

Targeted Instruction

Instruction aligns to specific student needs and learning goals

Student Reflection and Ownership

Ongoing student reflection promotes ownership of learning

Data Driven Decisions

Frequent data collection informs instructional decisions and groupings

New Courses & New Ecosystem

(Pratikno, 2018)

INDUSTRIAL REVOLUTION 4.0
(Digitization, Computing Power, Data Analytics)

Technology:
- Cyber-physical
- Internet of Things
- Biology

Application:
- Agricultural
- Logistics
- Manufacturing
- Etc

Science & Tech Fac

SocHum

Content

DISRUPTION:
(Uncertain, Unpredictable Future; Constant Change)

The Deep Shift:
- Physical → Virtual
- Human → Automation & Robotics
- Intermediary → Disintermediary

The Paradox:
- Voice vs Surveillance
- Empowerment vs Agglomeration
- Jobs Lost vs Jobs Gained
- Abundance Scarcity

SciTech Fac

Soci & Humanity Fac

Content

BE THE CHAMPION:
(Disruptor instead of disrupted, taking benefit from unpredictable future; Drive the future with empathy)

Educ: Powerful Agile Learner:
- Empathy
- Curiosity & Creativity
- Complex Problem Solving
- Determination

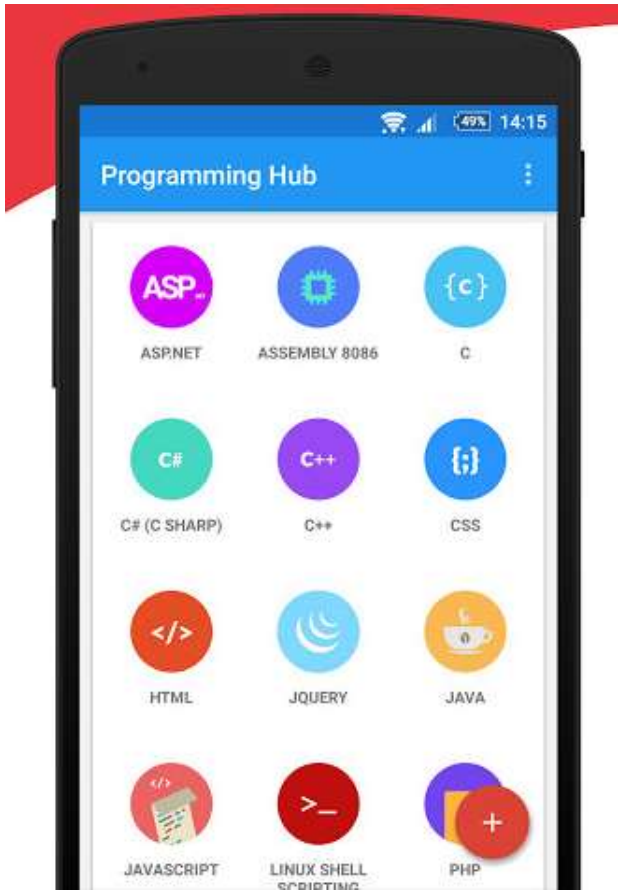
Rese: Visionary Contribution:
- Empathy
- Collaboration accros disciipline & Proffesion
- High end products

New Ecosystem:
- Collaboration
- Talent Pooling
- Goals oriented
- Collaboration & Sharing

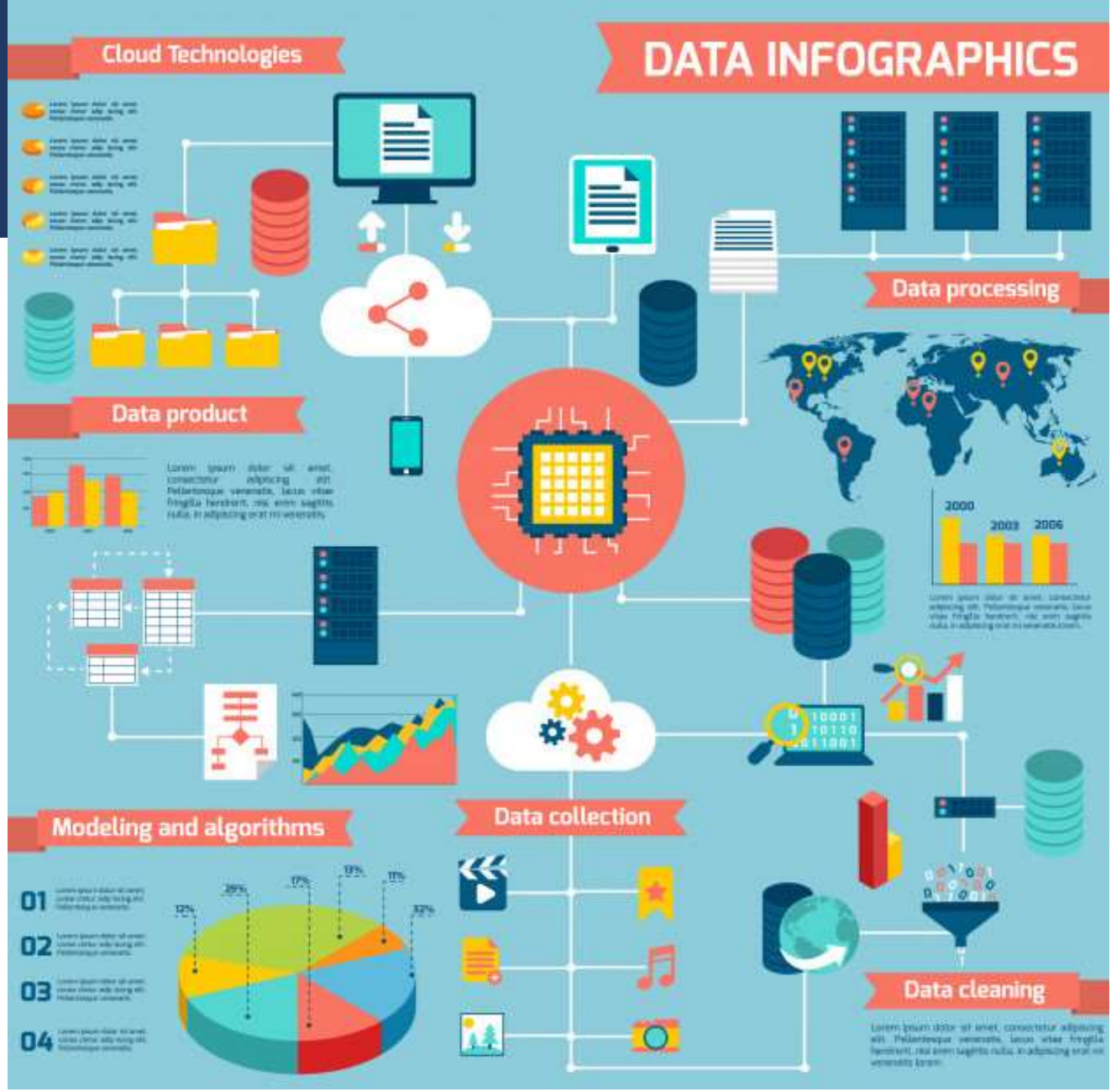
Character

Multi→Inter→Trans-Disciplinary

New ‘Compulsory’ Subjects



Programming/ Coding



Algorithm

Data Analytics

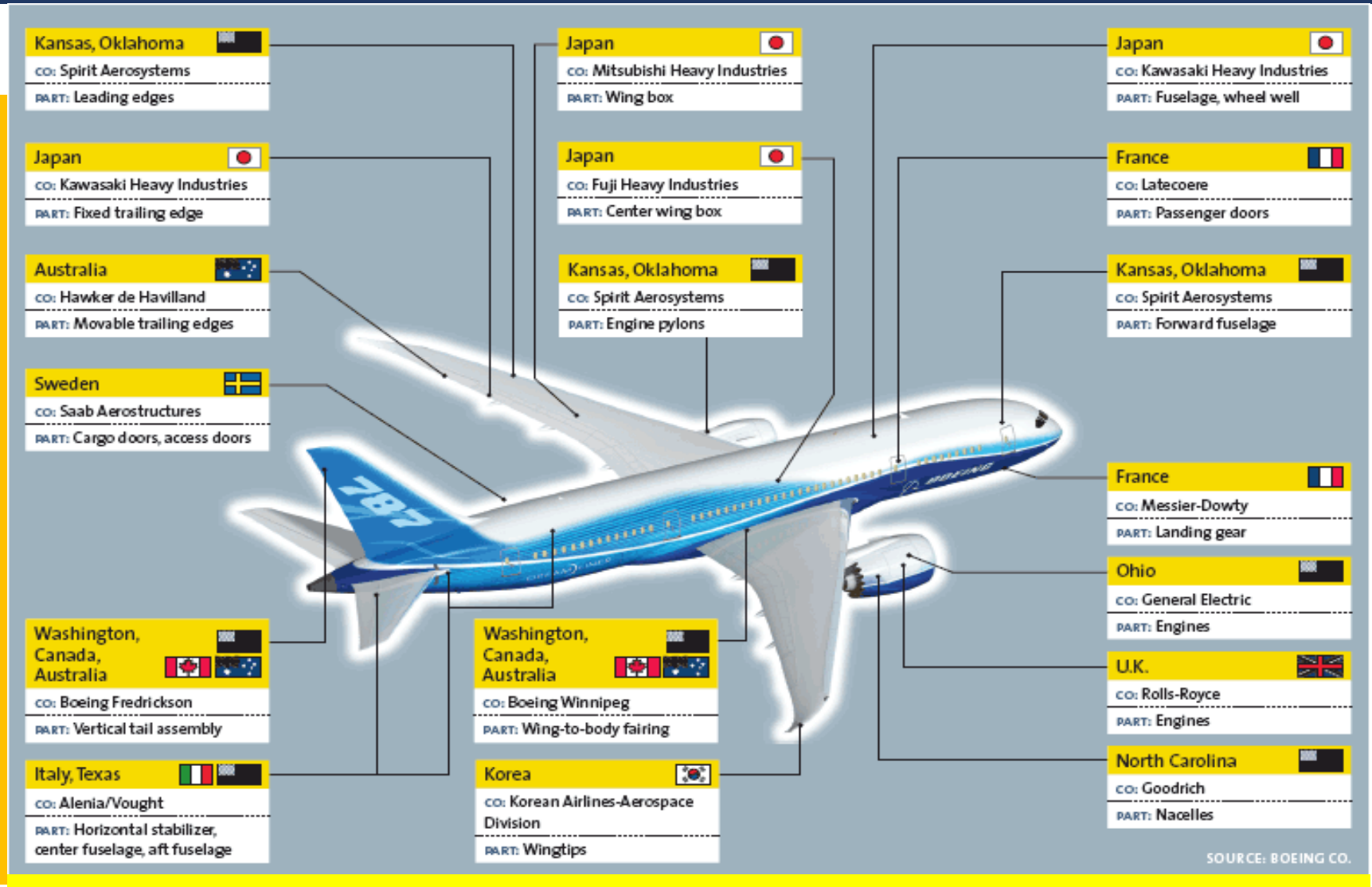
Organizational Ecosystem: Talent Pooling

- Platorm to attract top talents
- Scouting top talents, with specialized interests & high demand skills
- Networking
- Incentive & Recognition
- Collaboration



Organizational Ecosystem: Collaboration and Sharing

- Multi-Inter-Transdisciplinary Research & Learning
- Virtually connected
- Co-creation, shared resources, shared information



Boeing Global Collaboration

Powerful Agile Learner: Skillset Needed

10 TOP SKILLS NEEDED IN 2020
(that could not be replaced by automation)
(WEF, 2017)

1

COMPLEX PROBLEM SOLVING

To put it simply, it's about using analysis to make decisions and implement solutions.



2

CRITICAL THINKING

Increased complexity and sophistication of businesses in the coming years will mean more and more employers will be looking for critical thinking abilities in the candidates they interview.



3 Creativity

4 People management

5 Coordinating with others

6 Emotional Intelligence

7 Judgement and decision making

8 Service orientation

9 Negotiation

10 Cognitive flexibility

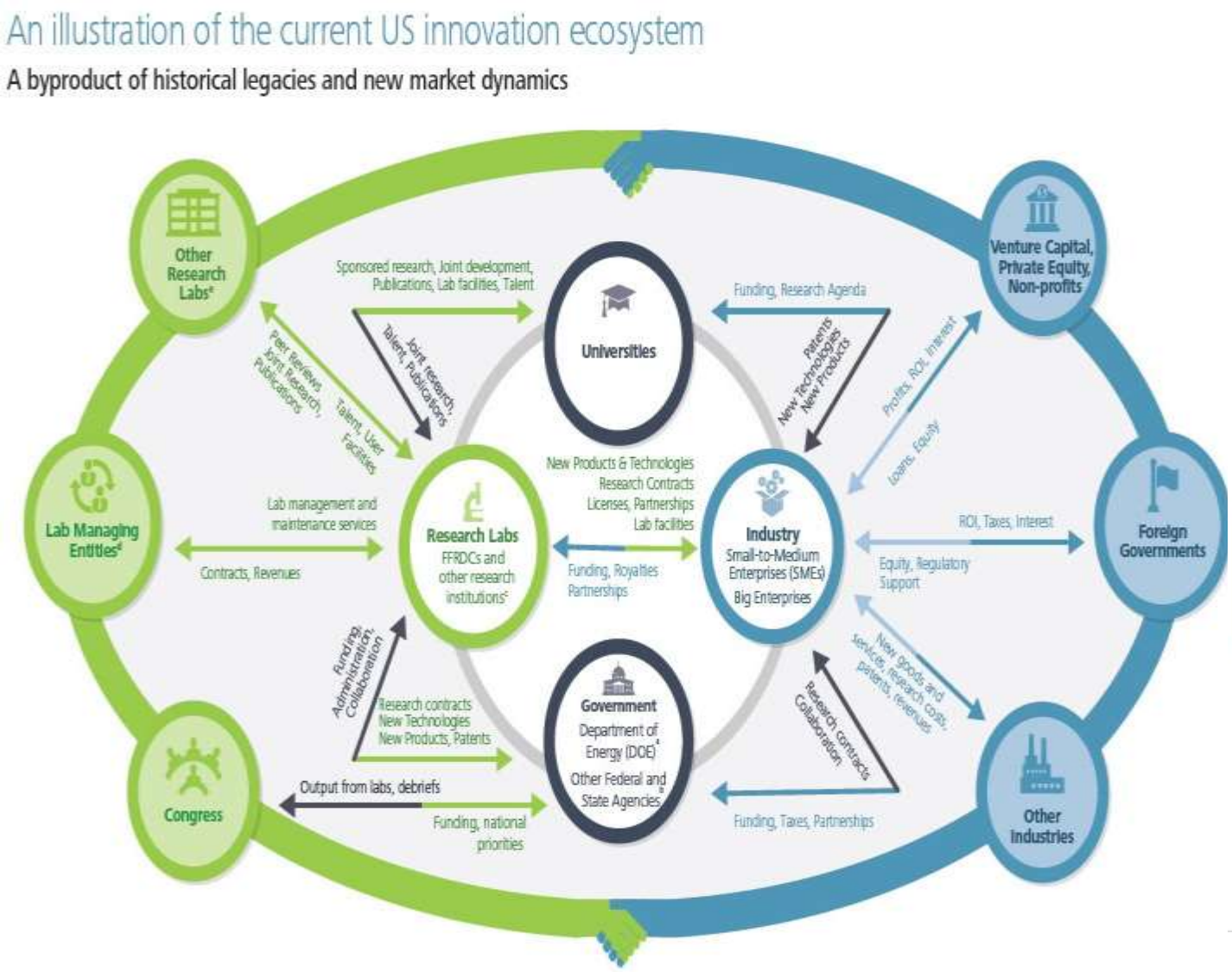
WORLD ECONOMIC FORUM

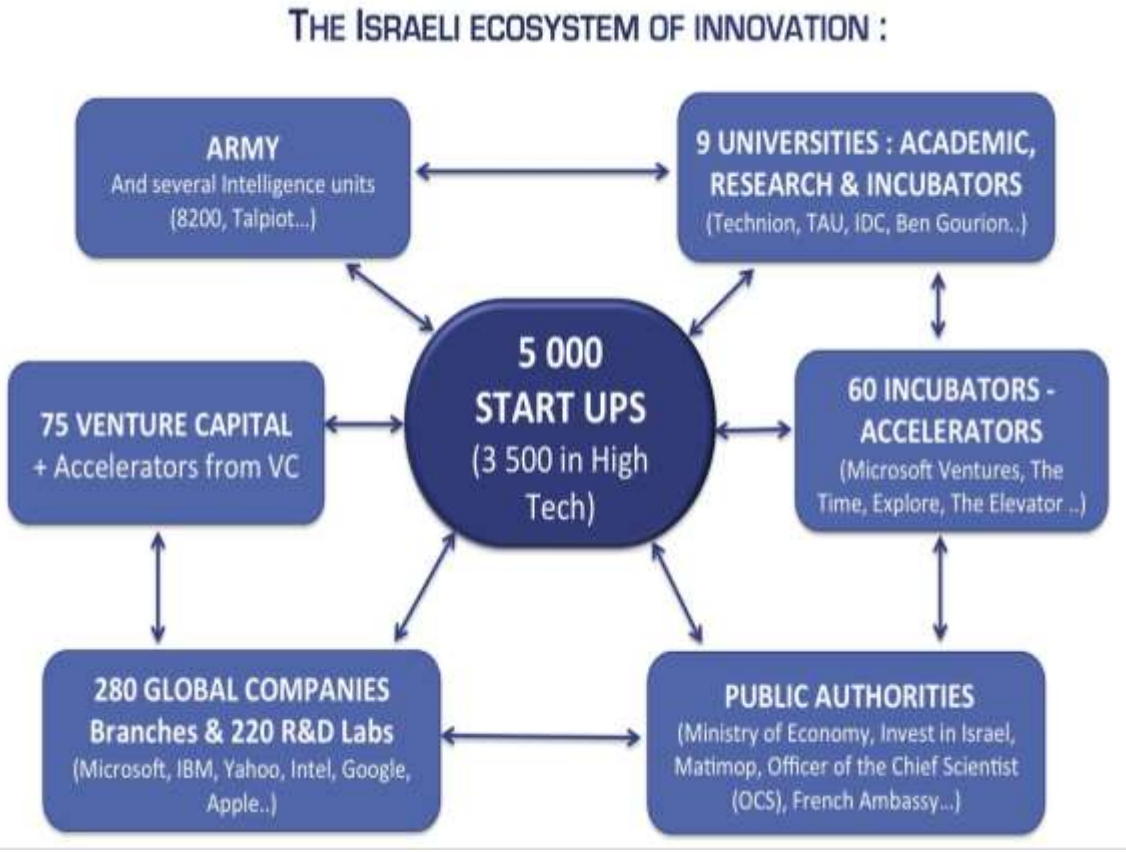
Powerful Agile Learner: Curiosity, Creativity, Determination



“Eksperiment” from Nicholas Negroponte:
Founder of One Laptop per Child; Co-Founder of Media Lab MIT

Ecosystem 5.0: Connecting the dots…





Walk the Talk: It all depends on VISIONARY Leadership

- Foresee the future
- Challenge the process – be a pioneer
- Be enthusiastic – inspire others
- Help others to act – Not a consensus for all, but build **COALITION**



Walk the Talk: Self Disrupt

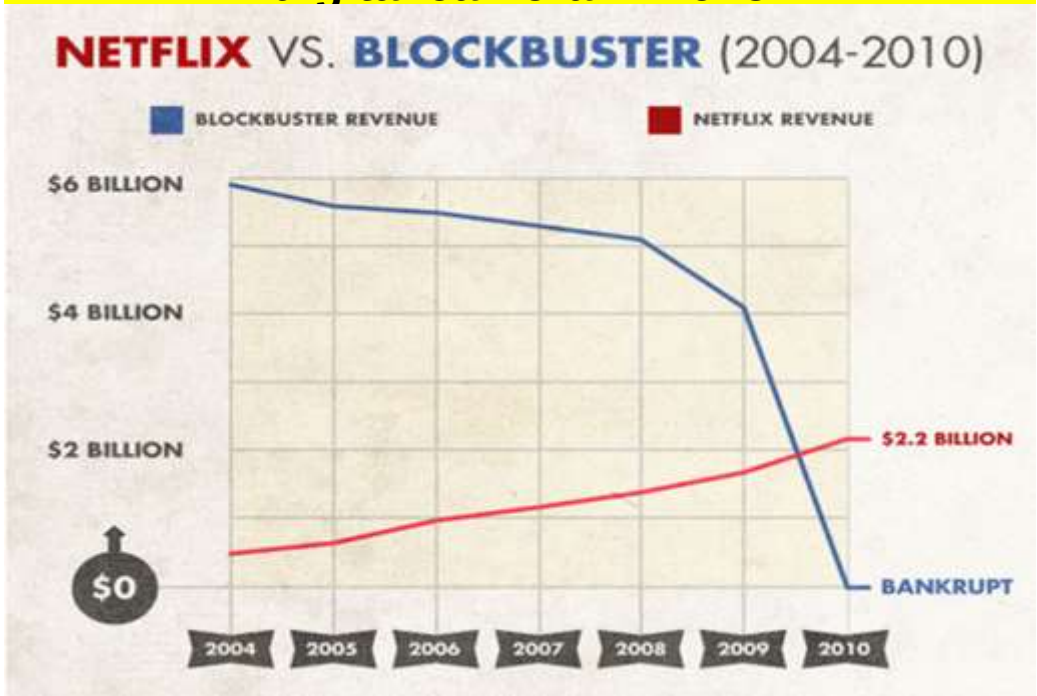
- If you don’t **innovate**, others will do
- If you don’t **disrupt**, others will do

- Be a **disruptor,** otherwise you’ll be disrupted
- Develop **internal disruptor,** before others disrupt you

You need a powerful
disruptive intelligence



Steve Sasson: the inventor of Kodak's first digital camera in 1975





@KemensetnegRI



kemensetneg.ri



GOALS

Be Disruptor instead of disrupted – taking benefit from unpredictable future; Drive the future with empathy

Pendidikan

Penelitian & Pengabdian

Powerful Agile Learner:

- Empathy *(kemanusiaan & kemajuan bangsa)*
- Curiosity & Creativity
- Complex problem solving
- Determination

Visionary Contribution:

- High-end products
- Relevant research & solving problems

STRATEGY

Dynamic model that ignites passion



- Shared purpose & vision
- Sensing and seizing opportunities
- Flexible resource allocation
- Actionable strategic guidance

PEOPLE

Embodied dreams across the organization



- Cohesive community
- Shared & visionary leadership
- Entrepreneurial drive
- Role mobility

PROCESS

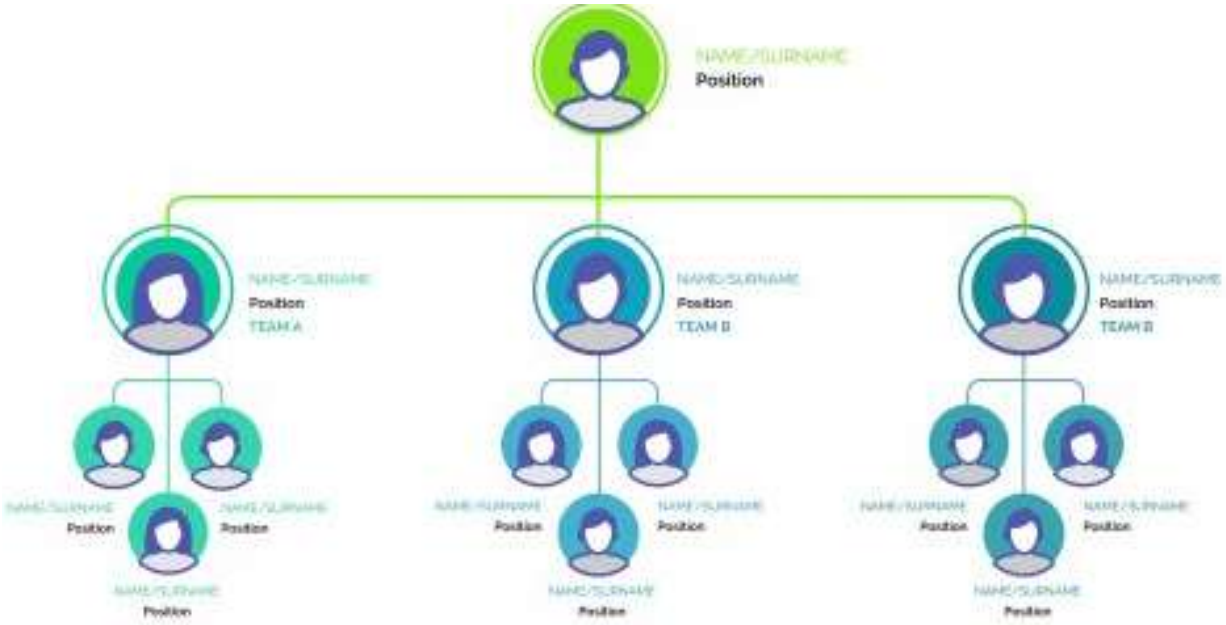
Rapid decision & learning cycles



- Rapid iteration & experimentation
- Standardized ways of working
- Performance orientation
- Information transparency
- Continuous learning
- Action-oriented decision making

STRUCTURE

Network of empowered teams



- Clear, flat structure
- Clear accountable roles
- Hands-on governance
- Robust communities of practice
- Active partnership & ecosystem
- Open physical & virtual environment
- Fit-for-purpose accountable cells

Next-generation enabling technology



- Evolving technology architecture, systems & tools
- Next-generation technology development & delivery practices